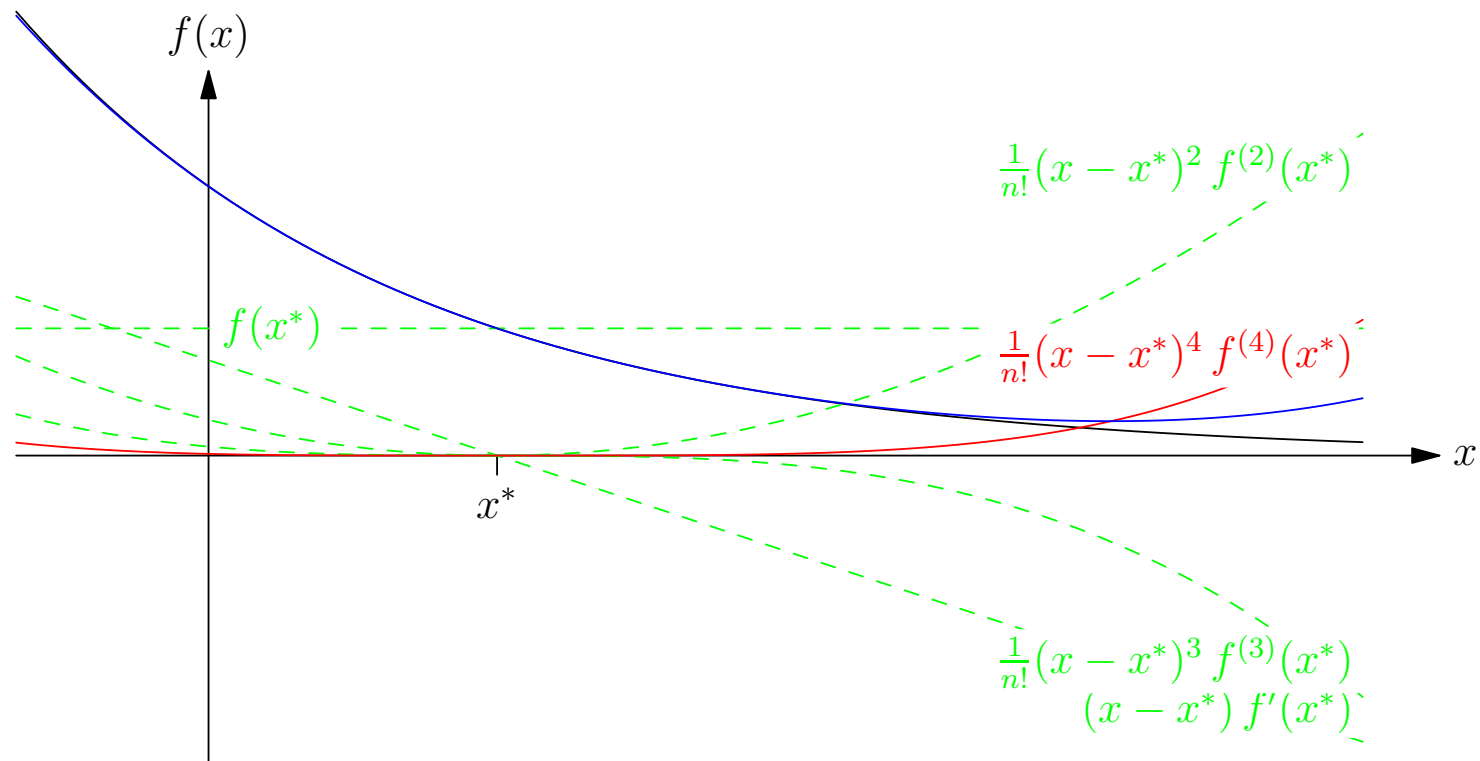


Mathematics of Machine Learning

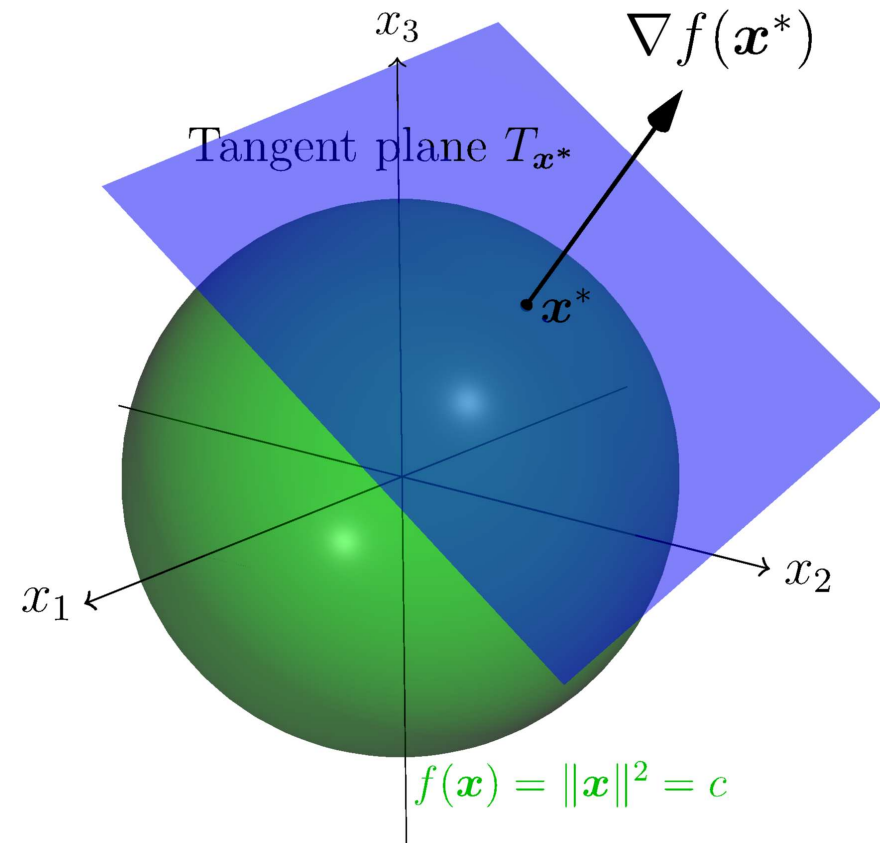
Taylor's Expansion



Convergence, remainder, high-dimensions, Hessian

Outline

1. **1-D Taylor Expansion**
2. High Dimensions
3. Hessians



Approximating Functions

- In much of machine learning we are trying to approximate functions■
- There are many different ways to do this■
- Depending on your application some are much better than others■
- All function approximation makes some assumptions about the regularity of the function they are approximating■
- Taylor expansions are polynomial approximations of a smooth function around a point■

Polynomial Approximation

- Suppose we wanted to approximate a function $f(x)$ around a point x^* using a polynomial

$$f(x) \approx a_0 + a_1(x - x^*) + a_2(x - x^*)^2 + \dots + a_n(x - x^*)^n$$

- Taking derivatives

$$f'(x) \approx a_1 + 2a_2(x - x^*) + 3a_3(x - x^*)^2 + \dots + na_n(x - x^*)^{n-1}$$

$$f''(x) \approx 2a_2 + 6a_3(x - x^*) + \dots + n(n-1)a_n(x - x^*)^{n-2}$$

- One way to decide on the coefficients is to set $x = x^*$ and assume equality

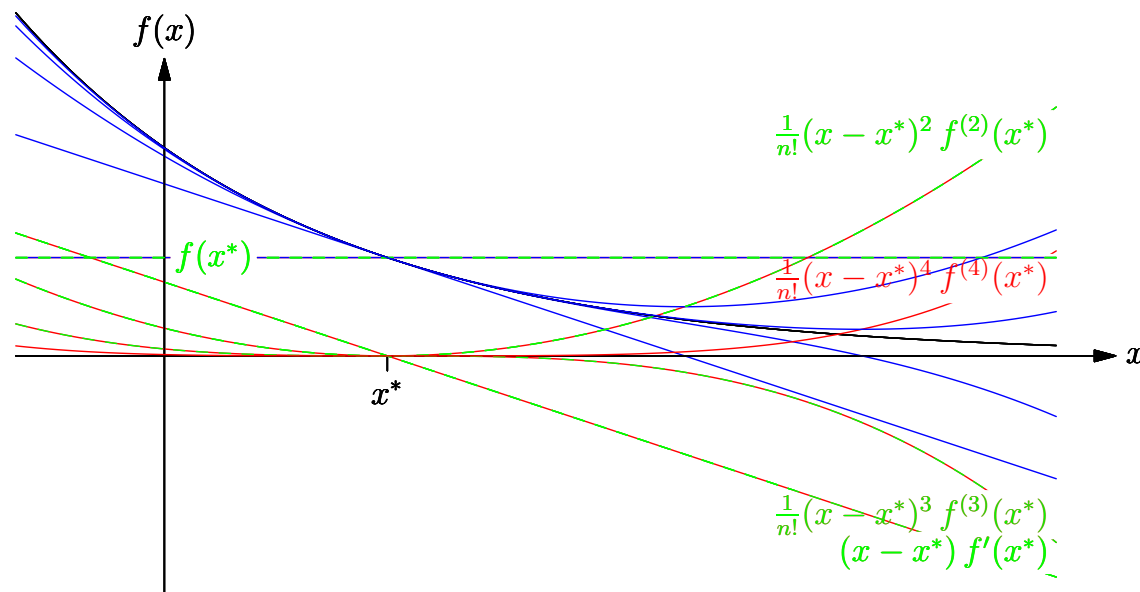
$$a_0 = f(x^*), \quad a_1 = f'(x^*), \quad a_2 = \frac{1}{2}f''(x^*), \quad a_i = \frac{1}{i!}f^{(i)}(x^*)$$

Taylor Expansion

- This gives the Taylor expansion

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*) + \frac{1}{2}f''(x^*)(x - x^*)^2 + \dots + \frac{1}{n!}f^{(n)}(x^*)(x - x^*)^n$$

- The left and right-hand side both have the same n derivatives at the point x^*



Convergence

- As we add more and more terms our local approximation becomes more accurate, provided the higher derivatives exist
- Polynomials have the nice property that they only diverge at ∞
- However, polynomials cannot capture functions that diverge for finite numbers, e.g.

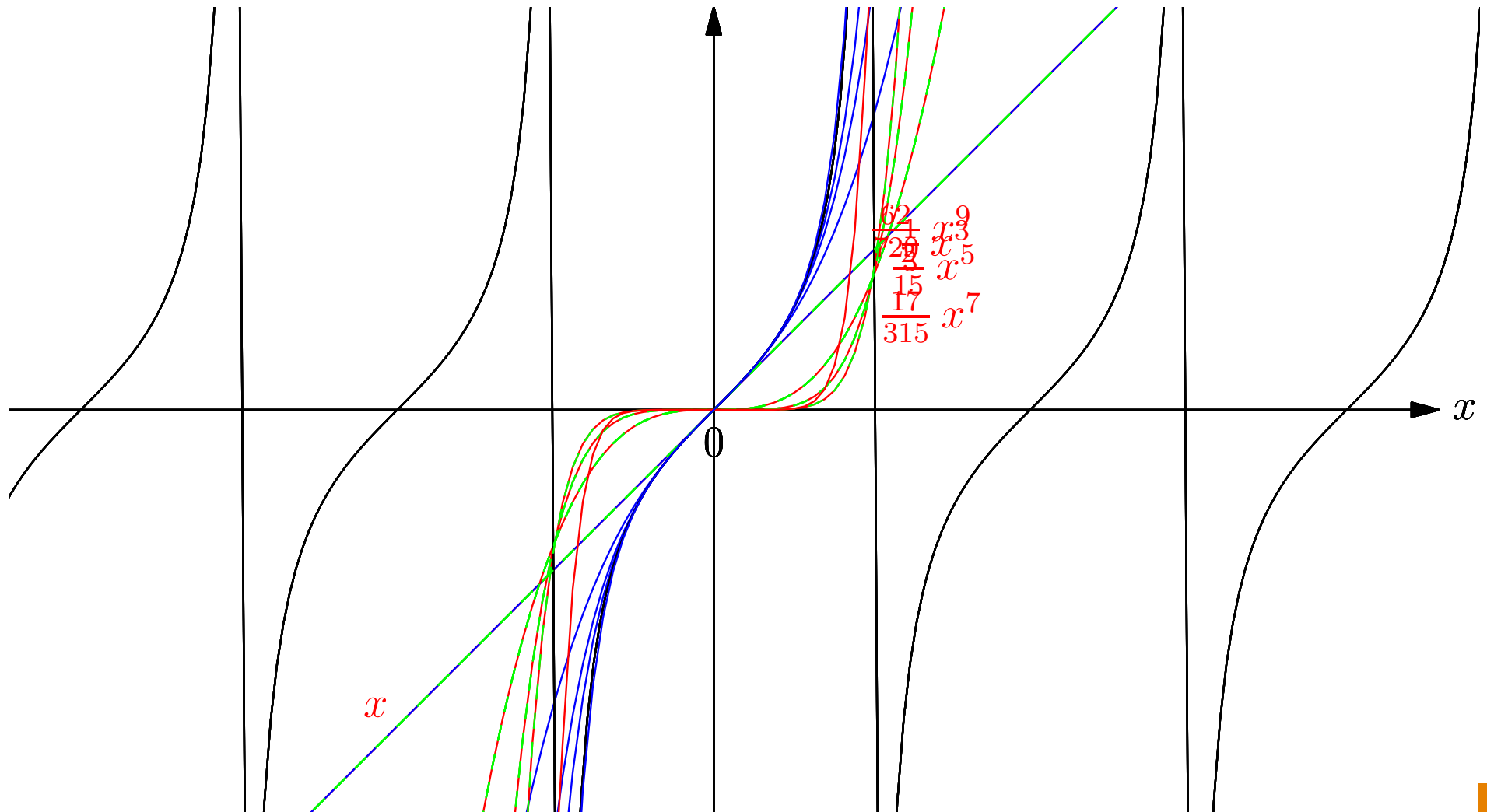
$$\frac{1}{x}$$

$$\frac{1}{(x - 5)^2}$$

$$\tan(x)$$

- This leads to a loss of global convergence

Taylor Expansion of $\tan(x)$



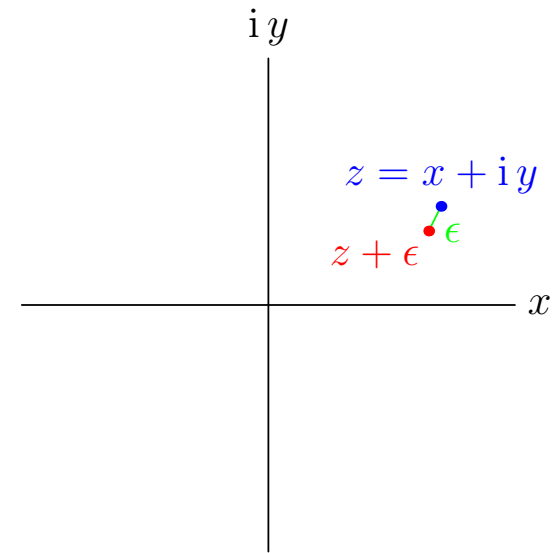
Complex Functions

- To understand convergence of Taylor series more generally we need to go into the complex plane■
- We can generalise many functions by considering them as a function with a complex argument $z = x + iy$ ■
- The value of the complex function will be complex valued■
- Complex functions are a huge and beautiful field (almost all of which go beyond this course)■

Analytic Functions

- An important class of functions are those where for almost all z the derivative

$$\frac{df(z)}{dz} = \lim_{\epsilon \rightarrow 0} \frac{f(z + \epsilon) - f(z)}{\epsilon}$$

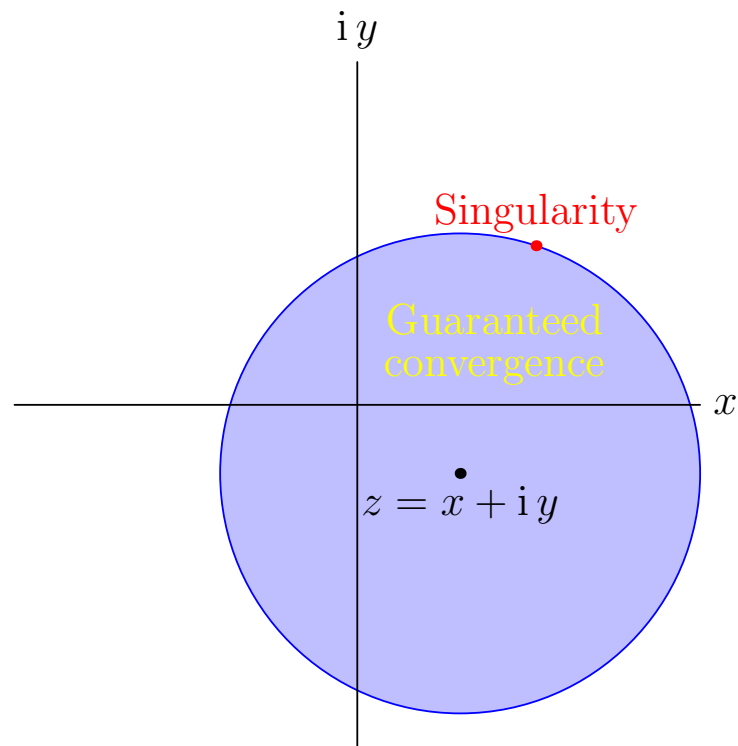


does not depend on the direction ϵ ■

- This includes polynomials, but also includes exponentials, trigonometric functions, logarithms, ratios of polynomials, “special functions”, etc. ■

Guaranteed Convergence of Taylor Series

- For analytic functions, Taylor series are guaranteed to converge inside a circle that excludes singular points

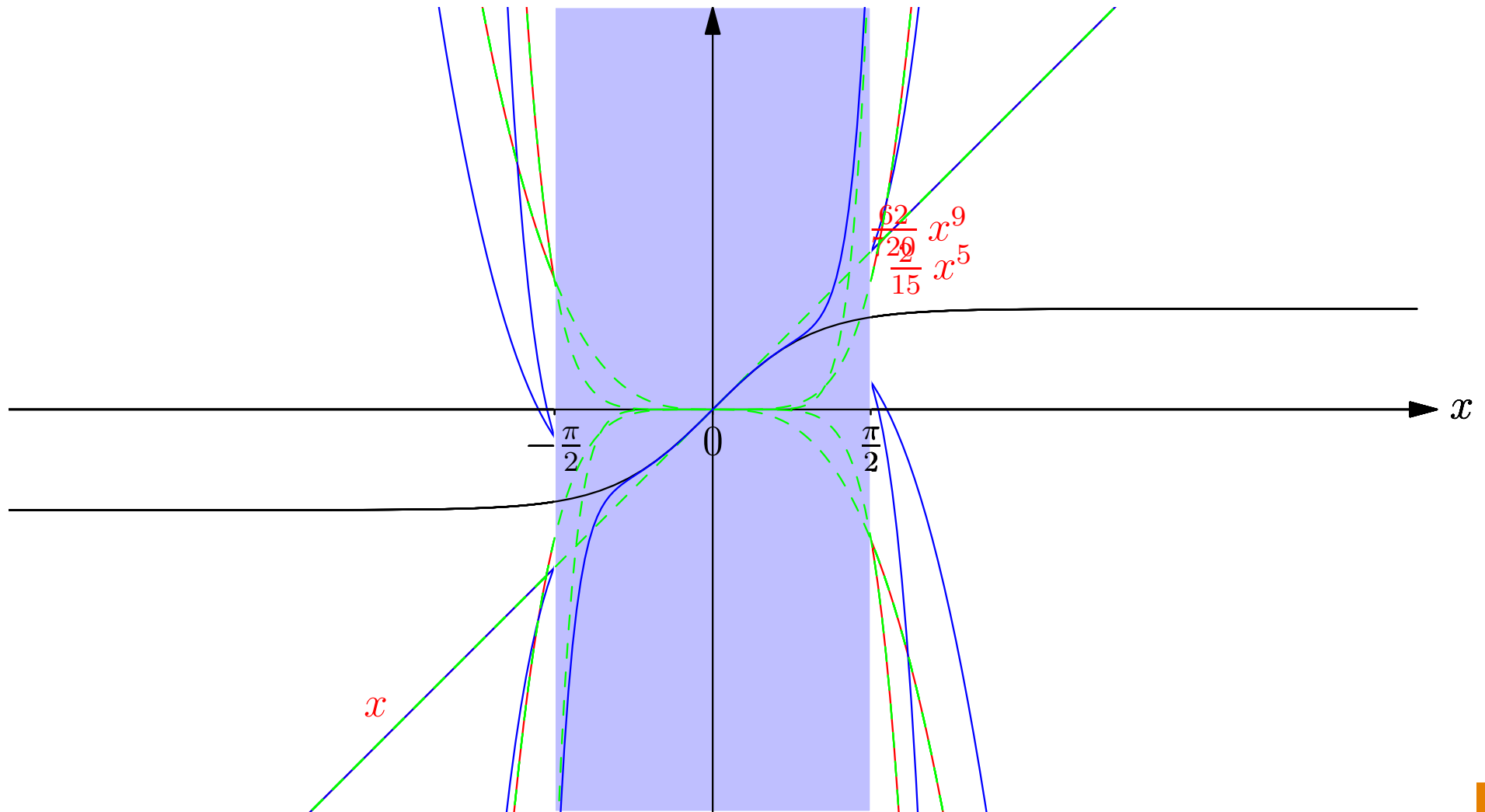


- E.g. the function $f(z) = \frac{1}{z^2+1}$ has a singularity at $z = \pm i$. The Taylor series of $f(z)$ around 0 is guaranteed to converge inside a circle of radius 1.

The Hyperbolic Tangent

- The hyperbolic tangent, $\tanh$, is related to the normal tangent function through $\tanh(x) = -i \tan(ix)$ ■
- The function (along the real line) doesn't have any singularities■
- But we know that $\tan(x)$ has singularities at $\frac{\pi}{2} + k\pi$ where k is any integer, so $\tanh(x)$ has singularities at $x = \pm i\frac{\pi}{2}, \pm i\frac{3\pi}{2}, \dots$ on the imaginary axis■
- Thus a Taylor expansion of $\tanh(x)$ around $x = 0$ is guaranteed to converge in a circle of radius $\pi/2$ ■ but not necessarily for larger values of x ■

Taylor Expansion of $\tanh(x)$



Remainder

- Taylor expansions actually come with an error bound

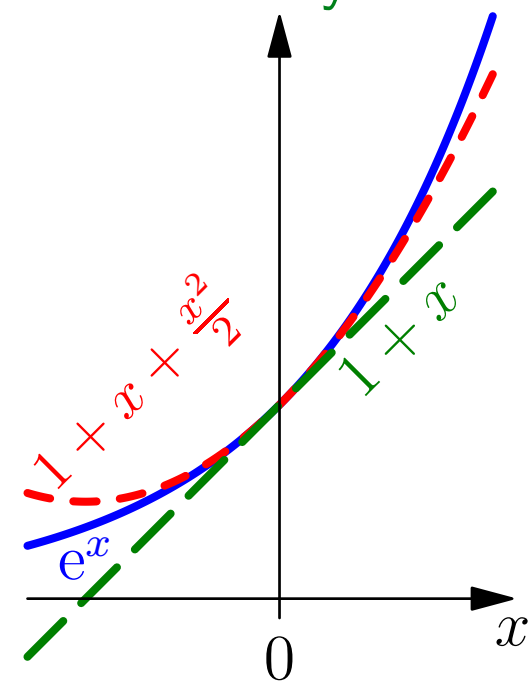
$$f(x) = \sum_{i=0}^{n-1} \frac{(x - x^*)^i}{i!} f^{(i)}(x^*) + \frac{(x - x^*)^n}{n!} f^{(n)}(\theta_x)$$

θ_x is a number in the interval between x^* and x ■

- The last term, known as the **remainder**, can often be very useful

$$e^x = 1 + x + \frac{x^2}{2} e^{\theta_x} \blacksquare$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} e^{\theta_x} \blacksquare$$



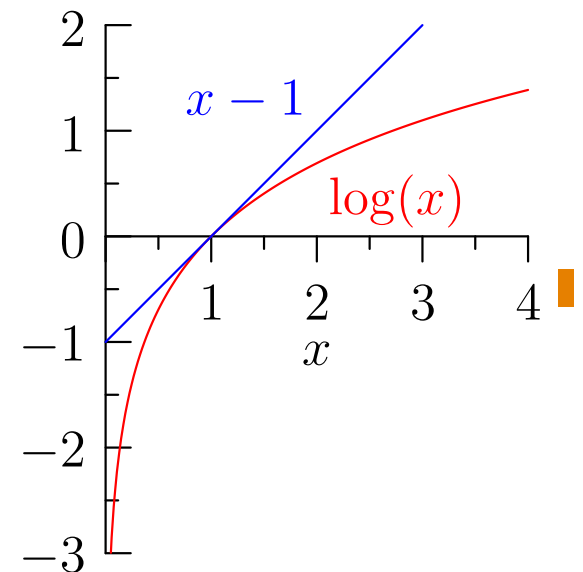
Useful Bounds

- A much used bound in mathematical analysis is $e^x \geq 1 + x$
- Many other bounds can be obtained similarly, e.g. expanding $\log(x)$ around $x = 1$

$$\log(x) = (x - 1) - \frac{(x - 1)^2}{2} \frac{1}{\theta^2}$$

giving

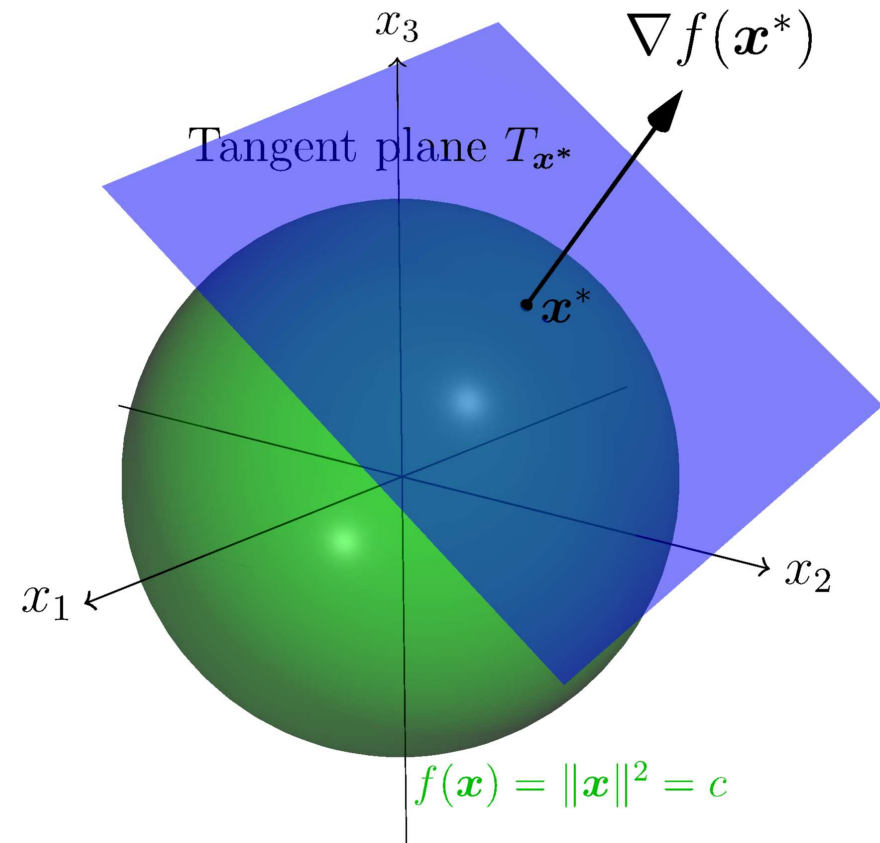
$$\log(x) \leq x - 1 \quad \text{or} \quad \log(1 + x) \leq x$$



- The proof of the remainder term comes from applying multiple integrals to the term $f^{(n)}(x)$. For the sake of time I will skip this

Outline

1. 1-D Taylor Expansion
2. **High Dimensions**
3. Hessians



High Dimensional Spaces

- Just as we defined a Taylor expansion by approximating the function by a polynomial we can do the same for a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$

$$f(\mathbf{x}) \approx c^{(0)} + \sum_{i=1}^n c_i^{(1)} (x_i - x_i^*) + \sum_{i,j=1}^n (x_i - x_i^*) c_{ij}^{(2)} (x_j - x_j^*) + \dots$$

- We can determine the coefficients by differentiating both sides and setting $\mathbf{x} = \mathbf{x}^*$

$$f(\mathbf{x}^*) = c^{(0)}$$

$$\frac{\partial f(\mathbf{x}^*)}{\partial x_i} = c_i^{(1)}$$

$$\frac{\partial^2 f(\mathbf{x}^*)}{\partial x_i \partial x_j} = 2c_{ij}^{(2)} \quad (\text{choosing } c_{ij}^{(2)} = c_{ji}^{(2)})$$

High Dimensional Taylor Expansion

- We find

$$f(\mathbf{x}) \approx f(\mathbf{x}^*) + (\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^\top \mathbf{H}(\mathbf{x} - \mathbf{x}^*) + \dots$$

- Where the matrix $\mathbf{H}$ is known as the Hessian and has components

$$H_{ij} = \frac{\partial^2 f(\mathbf{x}^*)}{\partial x_i \partial x_j}$$

- Because

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = \frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i}$$

the Hessian is symmetric ($\mathbf{H} = \mathbf{H}^\top$)

Tangent Plane

- For smooth functions in n dimensions the equation $f(\mathbf{x}) = c$ defines an $(n - 1)$ -dimensional hyper-surface (because we have imposed one constraint)
- This can be thought of as the high dimensional analogue of a contour line
- The plane that goes through a point $\mathbf{x}^*$ that lines up with the contour surface is known as the tangent plane
- Away from $\mathbf{x}^*$ the value of the function $f(\mathbf{x})$ differs from the tangent plane by a quantity of order $\|\mathbf{x} - \mathbf{x}^*\|^2$

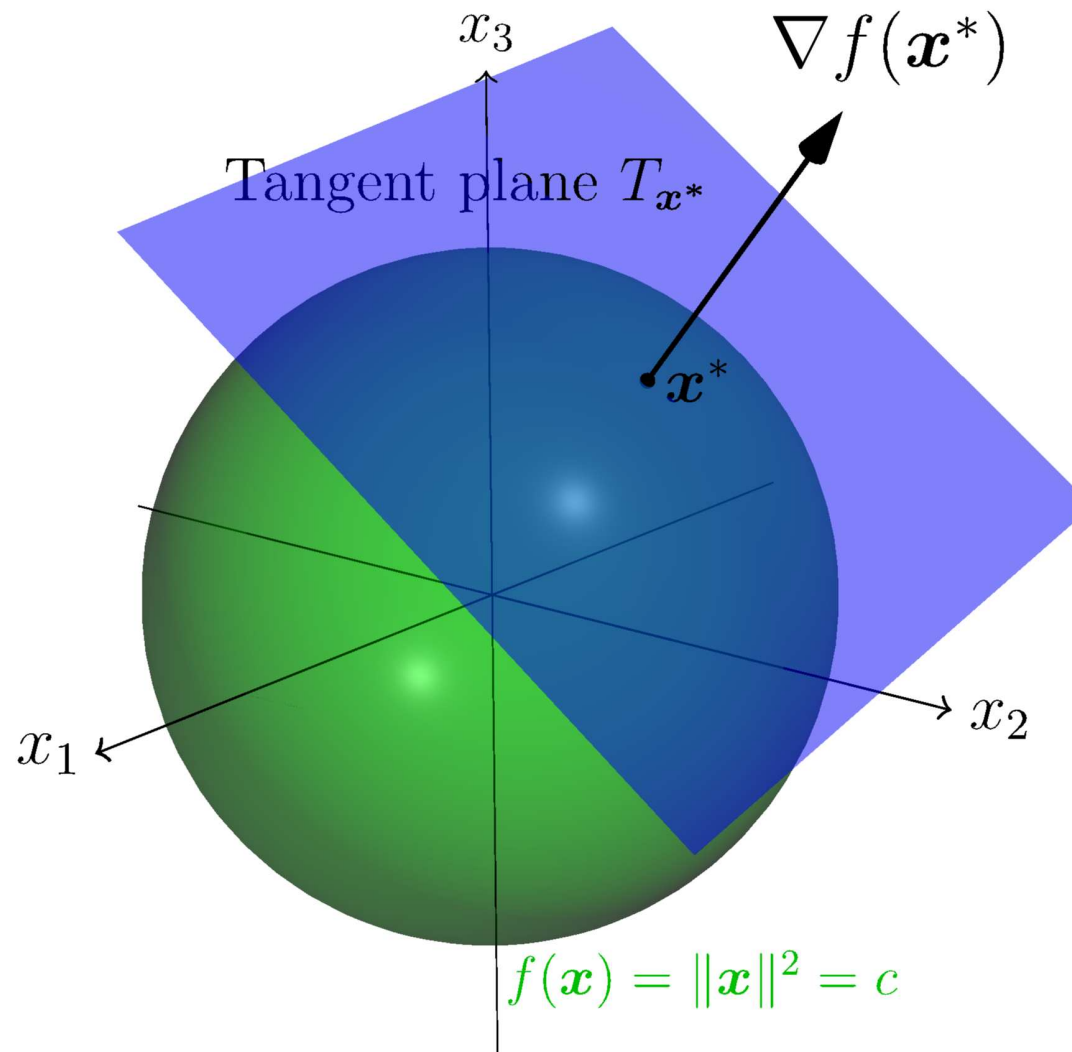
Defining the Tangent Plane

- Thus if $\mathbf{x}$ is in the tangent plane $T_{\mathbf{x}^*}$ then

$$f(\mathbf{x}) = f(\mathbf{x}^*) + O(\|\mathbf{x} - \mathbf{x}^*\|^2) \blacksquare$$

- Or $(\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) = 0 \blacksquare$
- That is the tangent plane at a point $\mathbf{x}^*$ is the set of points orthogonal to the gradient $\blacksquare$
- Notice that the gradient then points in the direction orthogonal to the tangent plane of the contour surface $\blacksquare$

Picture of Tangent Plane for $f(\mathbf{x}) = \|\mathbf{x}\|^2 = c$



Direction of Steepest Ascent

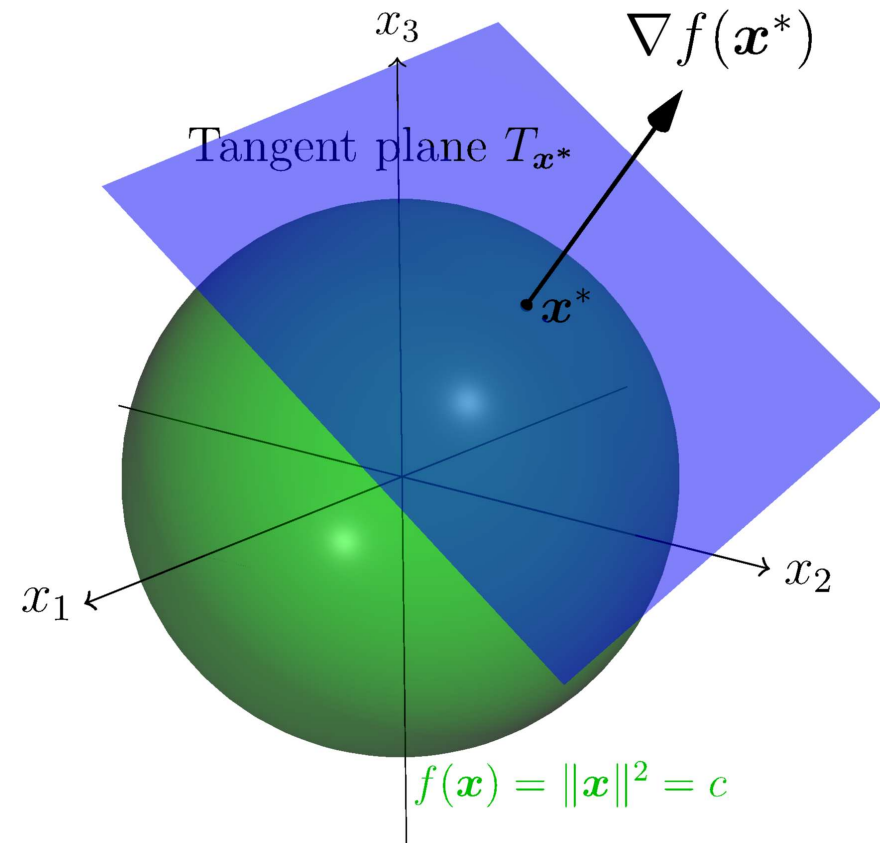
- We note that as

$$f(\mathbf{x}^* + \epsilon \mathbf{u}) = f(\mathbf{x}^*) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}^*) + O(\epsilon^2) \blacksquare$$

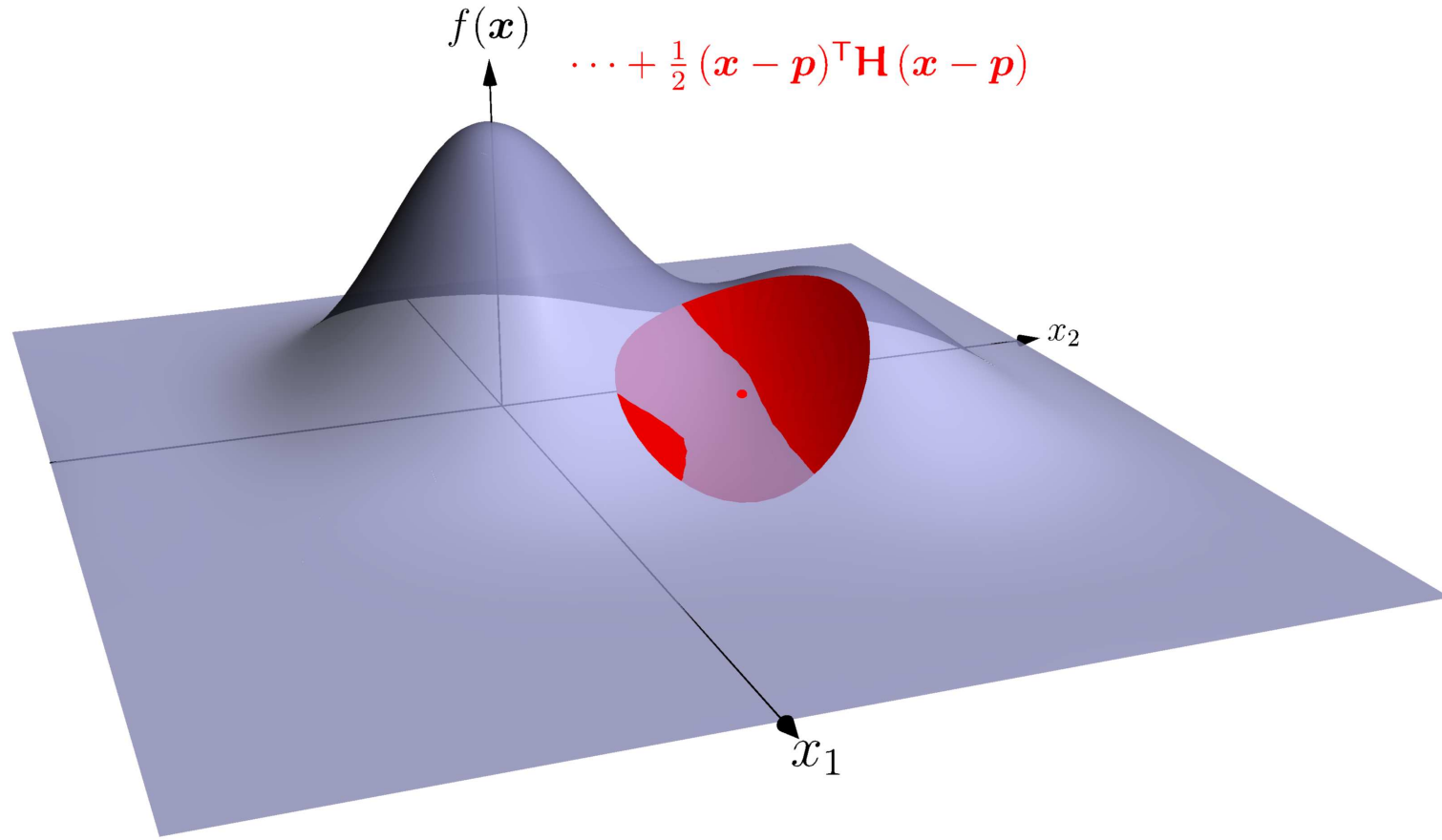
- For a step $\mathbf{u}$ of fixed length, the function locally grows fastest when $\mathbf{u}$ is in the direction $\nabla f(\mathbf{x}^*)$ $\blacksquare$
- Likewise when $\mathbf{u}$ is in the direction $-\nabla f(\mathbf{x}^*)$ the function decreases as fast as possible $\blacksquare$
- This is only true locally (when $\epsilon \gg \epsilon^2$) $\blacksquare$

Outline

1. 1-D Taylor Expansion
2. High Dimensions
3. **Hessians**



2-D Taylor Expansion



The Hessian

- The second order term in the Taylor expansion tells us about the **curvature** of the function■
- This is very relevant when we are at a **stationary point**, $\mathbf{x}^*$, where $\nabla f(\mathbf{x}^*) = \mathbf{0}$ so that

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^\top \mathbf{H}(\mathbf{x} - \mathbf{x}^*) + O(\|\mathbf{x} - \mathbf{x}^*\|^3)■$$

- If we make a perturbation $\mathbf{x} = \mathbf{x}^* + \epsilon \mathbf{u}$ then

$$f(\mathbf{x}^* + \epsilon \mathbf{u}) = f(\mathbf{x}^*) + \frac{1}{2}\epsilon^2 \mathbf{u}^\top \mathbf{H} \mathbf{u} + O(\epsilon^3)■$$

- For sufficiently small perturbation the sign of change in the function depends on the sign of $\mathbf{u}^\top \mathbf{H} \mathbf{u}$ ■

Eigenvalues

- We will revisit eigen-systems, but recall that for a square matrix an eigenvector, $\mathbf{v}_i$, of a matrix $\mathbf{H}$ is defined such that

$$\mathbf{H}\mathbf{v}_i = \lambda_i \mathbf{v}_i$$

- For an $n \times n$ symmetric matrix (such as the Hessian), the eigenvalues are all real, and we can choose n orthonormal eigenvectors such that $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij} = \mathbb{I}[i = j]$
- We note that

$$\mathbf{v}_i^\top \mathbf{H} \mathbf{v}_i = \mathbf{v}_i^\top \lambda_i \mathbf{v}_i = \lambda_i \mathbf{v}_i^\top \mathbf{v}_i = \lambda_i$$

- If we perturb in a direction $\mathbf{v}_i$ then

$$f(\mathbf{x}^* + \epsilon \mathbf{v}_i) = f(\mathbf{x}^*) + \frac{1}{2} \epsilon^2 \lambda_i + O(\epsilon^3)$$

General Directions

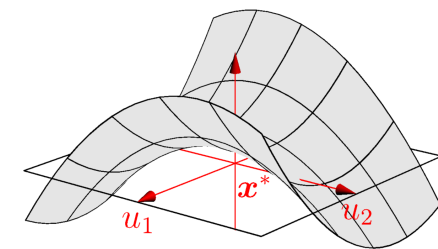
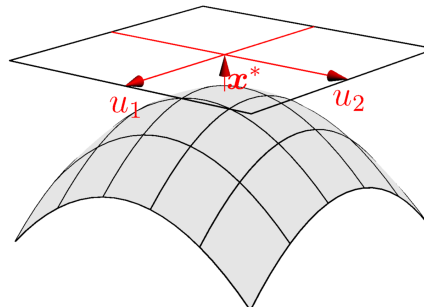
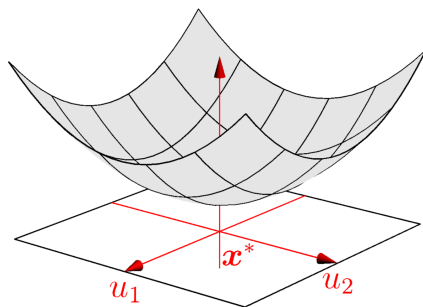
- We can write any vector $\mathbf{u}$ as a weighted sum of the eigenvectors (the eigenvectors form a basis) $\mathbf{u} = \sum_{i=1}^n c_i \mathbf{v}_i$ ■
- Note that

$$\begin{aligned}\mathbf{u}^\top \mathbf{H} \mathbf{u} &= \left(\sum_{i=1}^n c_i \mathbf{v}_i^\top \right) \mathbf{H} \left(\sum_{j=1}^n c_j \mathbf{v}_j \right) \text{ ■} \\ &= \left(\sum_{i=1}^n c_i \mathbf{v}_i^\top \right) \left(\sum_{j=1}^n c_j \lambda_j \mathbf{v}_j \right) \text{ ■} \\ &= \sum_{i,j=1}^n \lambda_j c_i c_j \mathbf{v}_i^\top \mathbf{v}_j \text{ ■} = \sum_{i=1}^n \lambda_i c_i^2\end{aligned}$$

using the orthogonality of eigenvectors $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij}$ ■

Minima, Maxima and Saddle-Points

- If all the eigenvalues are positive, $\lambda_i > 0$, then for any u we have $u^T H u > 0$ and x^* would be a local minimum. We say H is **positive definite** or $H \succ 0$.
- Similarly if all the eigenvalues are less than zero (H is **negative definite**, $H \prec 0$), then x^* will be a local maximum.
- If there are both positive and negative eigenvalues then x^* is a saddle-point (if some eigenvalues are zero we need higher order terms to decide).



Finding Optima

- We are often given a loss function, $L(\mathbf{w})$, which we wish to minimise
- A sufficient condition for $\mathbf{w}^*$ to be a local minimum is that $\nabla L(\mathbf{w}^*) = \mathbf{0}$ and $\mathbf{H} \succ 0$ where $\mathbf{H}$ is the Hessian of $L(\mathbf{w})$ at the point $\mathbf{w}^*$
- Except for a few very simple loss functions, there are likely to be many (and often a huge number) of local minima
- Unless $L(\mathbf{w})$ is very simple we are unlikely to be able to solve $\nabla L(\mathbf{w}) = \mathbf{0}$ analytically

Gradient Descent

- We usually proceed iteratively, starting from some $\boldsymbol{w}^{(0)}$

$$\boldsymbol{w}^{(t+1)} = \boldsymbol{w}^{(t)} - r \nabla L(\boldsymbol{w}^{(t)})$$

- This moves us in the direction that (locally) minimises the loss function $L(\boldsymbol{w})$
- As we get closer to the minimum $\nabla L(\boldsymbol{w}^{(t)})$ typically becomes smaller and we take smaller steps
- However, many nasty things can happen, and we have to be careful in how we perform gradient descent
- We will revisit this when we look at optimisation

Lessons

- For functions that are smooth (at least almost everywhere) Taylor series are really useful in understanding the local behaviour of the function■
- The gradient defines the direction in which a function increases fastest■
- This will be orthogonal to the tangent plane, where locally the function is a constant■
- The Hessian (not to be confused with the Jacobian) is the matrix of second derivatives defining the curvature away from the tangent plane■